

DELBURNE AGCM, AB, CanadaWMO: **710140**Lat: **52.1828N**Lon: **113.1757W**Elev: **889**StdP: **91.09**Time Zone: **-7.00 (NAM)**Period: **08-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-28.7	-25.9	-32.3	0.2	-28.6	-29.2	0.3	-25.6	10.3	-4.4	9.3	-5.1	2.4	150	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	27.8	16.7	26.0	16.0	24.4	15.3	18.6	24.5	17.5	23.6	16.5	22.5	2.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.5	13.1	21.4	15.2	12.0	20.2	14.0	11.1	19.1	56.5	24.5	52.6	23.7	49.7	22.7	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.8	6.7	DB	-32.1	31.3	2.6	2.4	-34.0	33.0	-35.5	34.5	-36.9	35.8	-38.8	37.6	(4)
(5)			WB	-32.4	20.7	2.4	1.4	-34.2	21.7	-35.6	22.6	-37.0	23.4	-38.7	24.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.7	-8.3	-8.8	-3.2	3.5	10.5	14.2	16.8	16.1	11.4	4.5	-3.3	-9.7	(6)	
	DBStd	11.35	8.45	8.14	7.42	5.25	4.48	3.07	2.77	3.18	4.73	5.08	7.11	7.94	(7)	
	HDD10.0	2989	566	528	409	201	48	4	0	1	38	182	399	612	(8)	
	HDD18.3	5373	824	761	667	444	244	127	63	83	212	430	648	870	(9)	
	CDD10.0	693	0	0	0	6	64	131	212	189	79	10	1	0	(10)	
	CDD18.3	39	0	0	0	0	2	4	17	13	3	0	0	0	(11)	
	CDH23.3	588	0	0	0	3	41	58	212	193	79	2	0	0	(12)	
	CDH26.7	107	0	0	0	0	2	6	42	41	16	0	0	0	(13)	
	WSAvg	2.9	2.9	3.0	3.0	3.3	3.2	2.9	2.5	2.4	2.6	3.2	3.0	3.0	(14)	
(15) Precipitation	PrecAvg	471	22	15	21	28	53	88	90	59	35	22	21	17	(15)	
	PrecMax	575	73	37	46	67	104	145	178	112	76	55	60	34	(16)	
	PrecMin	341	2	3	5	6	19	26	26	17	2	5	4	4	(17)	
	PrecStd	63	14	8	10	14	22	28	36	20	20	13	13	9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.7	8.3	14.2	21.8	26.8	27.1	30.0	30.1	29.0	22.2	15.9	8.2	(19)	
		MCWB	3.5	3.6	6.4	9.5	12.7	15.3	18.4	16.2	14.1	11.5	7.9	3.1	(20)	
	2%	DB	6.7	5.7	10.7	18.5	24.3	25.0	27.8	27.4	25.7	17.7	11.3	5.2	(21)	
		MCWB	2.5	1.7	4.4	8.5	12.0	15.2	17.8	16.7	14.1	9.6	5.5	1.2	(22)	
	5%	DB	5.0	3.5	8.2	15.1	21.8	23.1	25.7	25.5	22.9	14.9	7.7	3.3	(23)	
		MCWB	1.3	0.3	3.2	6.4	11.2	14.2	17.2	16.5	13.3	7.9	3.2	-0.1	(24)	
	10%	DB	2.9	1.7	5.8	12.1	19.2	21.2	23.8	23.4	20.0	12.2	5.3	0.9	(25)	
		MCWB	-0.1	-0.8	1.9	5.0	10.1	13.3	16.3	16.0	12.0	6.5	1.8	-1.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.0	3.8	7.0	10.3	13.9	18.1	20.7	19.6	16.1	12.2	8.4	3.5	(27)	
		MCDB	7.9	8.0	13.3	20.9	23.8	24.1	26.3	25.3	24.7	20.3	14.9	7.3	(28)	
	2%	WB	2.8	1.9	4.9	8.5	12.7	16.4	19.0	18.4	14.9	10.3	5.6	1.6	(29)	
		MCDB	6.4	5.4	9.9	17.9	21.7	22.9	24.8	24.5	23.2	16.4	10.4	5.2	(30)	
	5%	WB	1.5	0.6	3.4	6.9	11.8	15.2	17.8	17.3	13.7	8.6	3.7	0.0	(31)	
		MCDB	4.7	3.1	7.9	13.7	20.2	21.1	23.9	23.5	21.6	14.3	7.2	2.8	(32)	
	10%	WB	0.0	-0.7	1.9	5.4	10.8	14.1	16.9	16.2	12.6	7.0	2.1	-1.6	(33)	
		MCDB	2.8	1.7	5.4	11.2	18.1	19.7	22.6	22.3	19.5	11.4	4.7	0.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	9.5	9.8	11.0	11.9	11.0	11.6	12.0	12.1	10.4	8.7	8.4	(35)	
		MCDBR	9.2	11.0	12.4	16.4	15.9	14.1	14.8	15.7	17.2	14.4	12.2	10.1	(36)	
		MCWBR	6.4	8.1	7.4	8.2	6.6	6.1	6.8	7.1	7.7	7.8	7.4	7.4	(37)	
	5% WB	MCDBR	8.9	10.6	12.0	15.4	13.8	12.3	13.3	13.7	15.6	14.2	11.7	9.8	(38)	
		MCWBR	6.5	7.9	7.4	7.8	6.0	6.0	6.8	6.9	7.4	7.7	7.3	7.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.262	0.269	0.286	0.322	0.339	0.346	0.346	0.359	0.309	0.281	0.252	0.243	(40)	
	taud		2.311	2.390	2.436	2.377	2.385	2.395	2.419	2.362	2.476	2.535	2.473	2.332	(41)	
	Ebn at Noon		751	867	919	916	913	906	899	866	881	847	780	717	(42)	
	Edh at Noon		61	76	89	107	112	112	108	108	85	65	51	50	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.09	2.13	3.52	4.73	5.75	6.10	6.42	5.23	3.65	2.18	1.13	0.81	(44)	
	RadStd		0.09	0.18	0.27	0.42	0.37	0.38	0.33	0.36	0.39	0.26	0.11	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (81 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page